

Mohamed Elsayed

647-451-4724 | [Email](#) | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

University of Toronto

Bachelor of Engineering in Engineering Science

Relevant Coursework: Data Structures and Algorithms (Python/C), Linear Algebra, Engineering Mathematics and Computation, Calculus I/II

Toronto, Ontario

Sep 2025 – May 2030

EXPERIENCE

Technical Lead & Head of Technology

Aug 2023 – Present

Al Mannar Charity

Luxor, Egypt

- Designed and maintain the charity's core production database managing beneficiary records, financial disbursements, and sensitive personal data.
- Develop and support backend systems and internal tools in Python using a relational database, ensuring data integrity and reliability.
- Own software architecture and ongoing technical decision-making supporting day-to-day operations.

Undergraduate Research Assistant

July 2025 – Aug 2025

American University in Cairo (AUC)

New Cairo, Egypt

- Built a microcontroller-based control system to automate a precision translation stage for optics experiments.
- Designed and tested motor control and magnetic field circuits and integrated embedded software in C++.

Backend Intern

July 2024 – Aug 2024

eVision

New Cairo, Egypt

- Engineered a scalable file-parsing system to automate data ingestion and processing.
- Collaborated with frontend, database, and QA teams in a cross-functional agile environment.

Software Developer & Technical Animator

Oct 2021 – July 2023

Self-Employed

Remote

- Contributed to collaborative game projects reaching **1.5B+ visits** and **250K+ concurrent players**.
- Implemented gameplay systems, technical animation, and real-time VFX using Lua, Python, Moon Animator, and Blender.
- Completed independent commission work delivering custom visual effects and animation systems to fit client requirements.

PROJECTS

LaTeX Notetaker Converter | *TypeScript, JavaScript, CSS, Python*

Dec 2025 – Present

- Built a web-based tool to convert handwritten, typed, and audio notes into structured LaTeX documents optimized for notation heavy coursework and technical writing.
- Implemented OCR, parsing and automation pipelines to generate compilable LaTeX with support for equations, figures, and modular document organization.

Structural Beam Simulation | *Python, MATLAB*

Nov 2025

- Built a general-purpose beam simulation engine for arbitrary cross-sections by computing section properties directly from geometry; generating deflection curves, and factor-of-safety statistics for arbitrary loads using numerical integration.
- Developed MATLAB stress/deformation visualizations, including 3D deflection animations under moving/distributed loads, and validated predictions against physical tests to improve accuracy.

TECHNICAL SKILLS

Programming Languages: Python, C, Java, JavaScript, TypeScript, SQL, MATLAB, HTML/CSS, Lua

Hardware & Electronics: Microcontrollers, Motor Control, Circuit Design, PCB Assembly, Sensors

Developer Tools: Git, VS Code

Libraries: NumPy, Matplotlib